

The Power of Vaccination

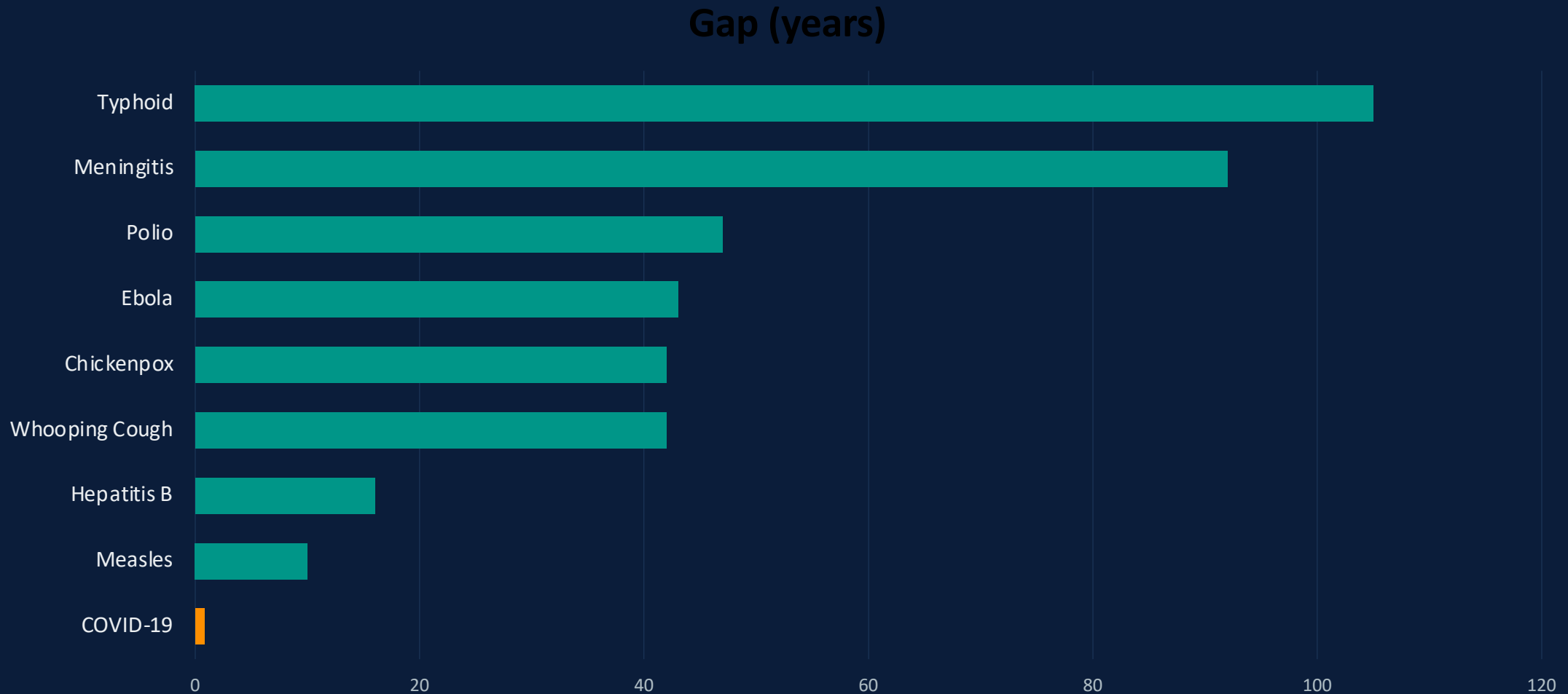
From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines have transformed public health over two centuries (1796 – 2020)

Source: Our World in Data · ourworldindata.org/vaccination

From 100+ Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen identification to vaccine licensure



COVID-19 highlighted — <1 year from identification to vaccine licensure, enabled by mRNA technology

100% Case Elimination Achieved for Diphtheria, Polio & Smallpox

Pre-vaccine vs. post-vaccine annual cases and deaths per million population in the United States | Source: Roush & Murphy, 2007

Disease	Pre-Vax Cases/M	Post-Vax Cases/M	Case Δ	Pre-Vax Deaths/M	Death Δ
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute Polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute Hep. B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

■ 100% reduction ■ <85% reduction — coverage gaps remain

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap reference: Measles cases per 100,000 across all 50 US states, 1929–2022

Pre-1963

1929–1962

100–1,000+ cases
per 100,000

Vaccine Era

1963–1979

Declining rapidly

Post-Mandate

1980–2022

Near zero
(sporadic outbreaks)

1963

First measles vaccine
(John Enders)

1971

MMR vaccine introduced
(Maurice Hilleman)

1980

Vaccination mandatory
for kindergarten entry

Incidence scale:



High (1,000+)



Moderate



Low



Near zero

Smallpox: The Only Human Disease Eradicated Through Vaccination

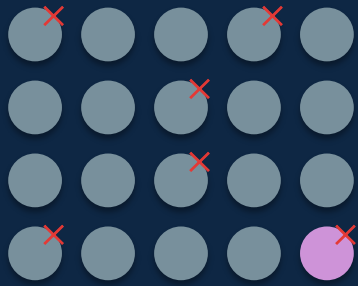


Why Smallpox Was Uniquely Suited for Eradication

- ✓ The virus infected only humans — no animal reservoir to sustain transmission
- ✓ Symptoms were clearly distinguishable — enabling rapid identification and isolation
- ✓ The vaccine was inexpensive, stable, and highly effective — enabling mass deployment

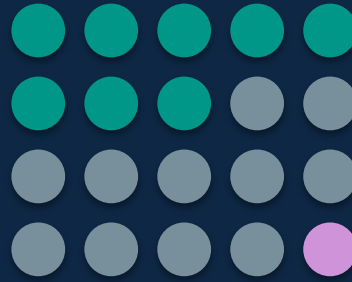
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

No Vaccination



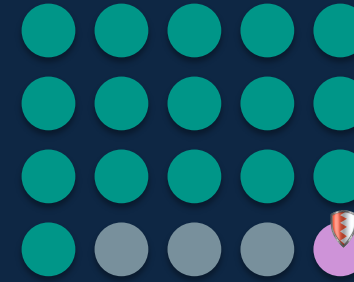
Pathogen spreads
freely through
population

Partial Vaccination



Some spread blocked
but vulnerable still
at risk

Herd Immunity



Enough vaccinated to
stop spread — vulnerable
are shielded



Vaccinated



Unvaccinated



Vulnerable (cannot be vaccinated)

Vaccines protect newborns, immunocompromised individuals, and cancer patients on chemotherapy who cannot be vaccinated.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

40.5%

Case reduction

31.3%

Death reduction

vs. 100% achieved for diphtheria, polio, smallpox

Hepatitis A

89%

Case reduction

88.7%

Death reduction

vs. 100% achieved for diphtheria, polio, smallpox

Acute Hepatitis B

83.9%

Case reduction

83.6%

Death reduction

vs. 100% achieved for diphtheria, polio, smallpox

Key Barriers to Closing Coverage Gaps

- ▶ **Access Gaps:** Not everyone globally has access to vaccines — supply chain and distribution limitations persist
- ▶ **Supply Shortages:** Vaccine supply shortages continue to put children at risk, especially in low-income settings
- ▶ **Trust & Uptake:** Poor scientific understanding reduces public trust and vaccine uptake in many communities

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1

100% Elimination

Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the United States.

2

Development Accelerated

Vaccine development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology.

3

Community Protection

Herd immunity shields the most vulnerable — newborns, immunocompromised patients, and those undergoing chemotherapy.

4

Coverage Gaps Remain

Pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%) still show incomplete reductions.

Call to Action for Global Health Systems

Expand access to vaccines in underserved regions · Close supply chain gaps to protect children at risk · Invest in public trust and scientific literacy · Leverage mRNA and genomic tools to tackle remaining diseases